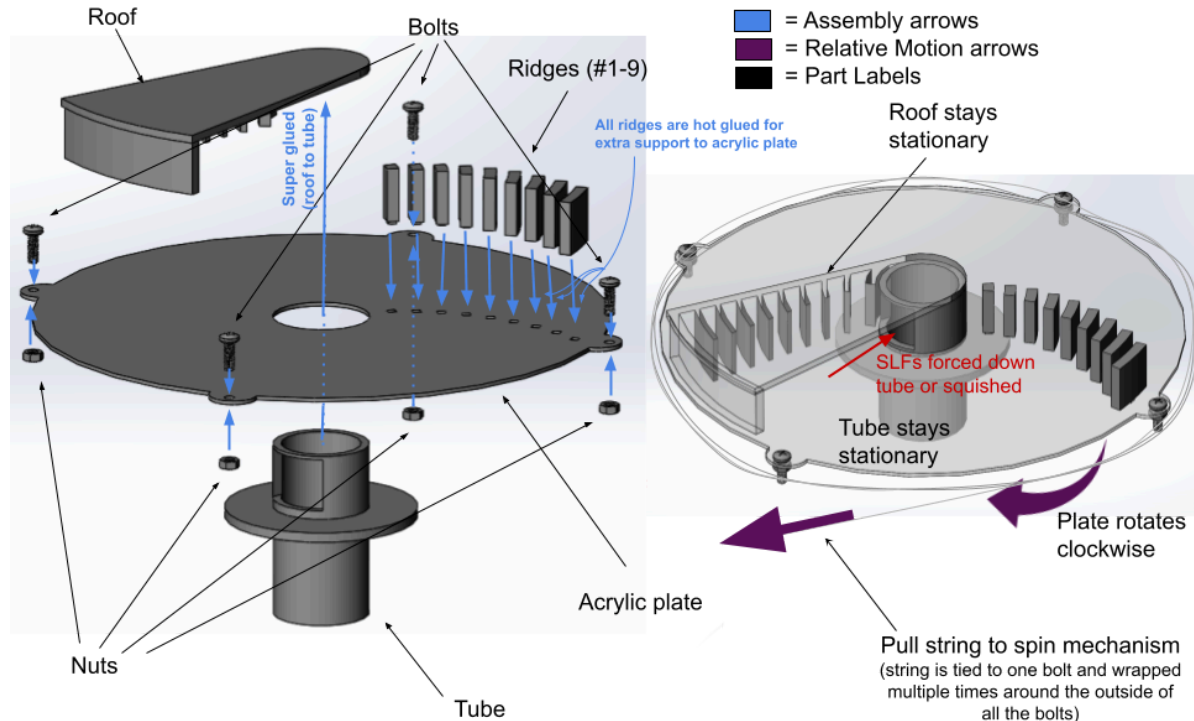


Functional Prototype

Exploded CAD view:



Part descriptions:

Acrylic Plate	Rotating disk that moves the ridges such that SLFs are put into scenarios where they must go down the tube, or be squished
Roof	Prevents SLFs from jumping off the acrylic plate once their set of ridges begins to force the SLFs down the tube
Ridges	Match up with ridges built into the roof and force SLFs down the tube, or squish SLFs that don't go down the tube. Prevents SLFs from getting away.
Tube	The component through which SLFs are forced down. Bottom opens into a containment zone where SLFs are imprisoned.
Nuts & Bolts	Used at the edges of the acrylic plate. String tied to one and wrapped around the rest multiple times → when the string is pulled, the plate spins.

Assembly instructions → combine components as shown in blue above (in exploded view); the only critical order of steps is that the acrylic plate must go down onto the tube before the roof is super-glued on (all other steps can happen in any order).

Critical dimensions → diameter of Acrylic plate = 9"; thickness of disk = 1/16"; height from disk to roof = 1"; height of ridges = 0.875"; Dimensions of tube opening = 1" x 1"

Design Documentation:

Part Name	Specs	McMaster Code	Fabrication
Ridges #1-9	0.875" tall, widths vary from .25"-1.25"	N/A	3d printed out of PLA and done at the RPL
Roof	r = 4.5", matching ridges 3d printed with roof as one piece	N/A	3d printed out of PLA and done at the RPL
Tube	1.5" diameter	N/A	3d printed out of PLA and done at the RPL
Clear Scratch- and UV-Resistant Cast Acrylic Sheet	12" x 12" x 1/16"	8560K171	Laser cut at RPL
Bolts	M4 12mm long from Taylor Design Studio	N/A	N/A
Nuts	M4 from TDS	N/A	N/A

Design Testing:

Light test: Does a sufficient percentage of light get through the clear acrylic compared to the amount inside? (ideally, at least 80% of light intensity passes through the acrylic so that the blocked light doesn't pose the risk of overheating the system)

Procedure: In a dark room, use a phone flashlight and another phone with a lumen reading phone app 5" apart. Repeat with the clear acrylic sheet in between while remaining 5" apart. Compare the values to identify the percentage of light that passes through.

Part Focus: Light and clear acrylic top

Results: We measured 740 lux coming from the phone without acrylic and 610 lux coming from the phone through the acrylic. This means that 83% of the light made it through the acrylic.

Conclusion: A sufficient proportion of light is escaping through the acrylic. No material change (in terms of thickness or composition) is needed for our next prototype.

Shake Test (Stability and Support): Does the trap maintain integrity during shaky conditions as might be expected on a windy day?

Procedure: Place the prototype in a desired location and shake it for 10s at 5 Hz (5 shakes per second), seeing if it continues to work during and after being shaken.

Part Focus: Entire prototype

Results: We shook it and it did not fall apart. However, the nuts on the four corners of the device used to support our spinning mechanism string came loose

Conclusion: The device is robust, but for our next prototype, we should use nylon insert nuts to ensure longevity under vibrational conditions.

Speed Test (Assembly and Fastening): Does our spinning top hold up when rotating at higher operating speeds?

Procedure: Spin the top as fast as possible. No part of the mechanism should break, and the top should still spin smoothly.

Part Focus: The spinning top

Results: No parts of the mechanism broke. We were able to spin it at upwards of 7.25 rad/s with no issue (spun 3 rotations in 2.6s). However, we did notice slight friction between the ridges on the disc and the roof. It did not appear to affect the rotation speeds – rotation speed remained constant through the friction occurring.

Conclusion: The device structure is robust and can operate at high speeds. However, we need to come up with a way of spinning it so we can get more consistent speeds (pulling a string is a bit inconsistent). Additionally, ensuring that ridges are vertical will eliminate any friction between them as they rotate.

Throw Tests (Load-bearing Performance): Will our device hold up when bugs land on it at high speeds?

Procedure: Fabricate or find a small object of the approximate mass of a lantern fly (0.5~1 inch). Toss it at varying degrees of speeds at the top of the device while spinning (~3m/s to match the jumping speed of SLFs). Neither the roof nor the spinning top plate should break. In addition, the bug should get “squished” by the rotation, though this criterion might not be met until our final prototype. We will try to get an object that isn’t rigid.

Part Focus: The ridges on the top of the device

Results: We threw a mix of 6 (0.5-inch) and 6 (1-inch) bug models at the top spinning device, and not only did it not break, it “caught” all bugs that landed. Additionally, the roof also remained sturdy under “landing impact” without issues.

Conclusion: The robust top against landing impact and spinning mechanism works well in either trapping the LSF or squashing it. This type of structure works well; no change needed here.

High Volume Test (Load Bearing and Performance): Can the trap handle more than one bug going down the shaft at one time?

Procedure: Try putting multiple lanternfly-shaped objects into the shaft hole at the same time. We approximate that 3-6 “bugs” will be able to fall through.

Part Focus: Bug shaft

Results: The 1 in bug models we made were too big to fit through the shaft. However, our 0.5 in models were able to be shaken out of the device. We were able to fit upwards of 6 0.5 in bugs into the assembly and still shake all of them out.

Conclusion: Due to the entrance to our tube being too similarly sized to a large SLF, they struggled to go down the tube when clumped. Therefore, for our next prototype, we would like to slightly increase the width of the opening (by approximately .25”).

Tolerance Test (Assembly and Fastening): Do our tolerances for the slotted ridges work out?

Procedure: Assemble the prototype. The ridges on the top should fit in tightly and not fall off.

Part Focus: Ridges

Results: Everything fit together. However, we had to use a little bit more superglue than we wanted to attach the ridges.

Conclusion: We concluded that we need a slightly more snug fit – 0.01” as measured by a caliper – to ensure that minimal glue is used. We will apply this to the next prototype.

Success Criteria:

Our project is a bug catcher that uses light during the evening to lure lanternflies into a spinning trap. The top consists of ridges that force the bug to either move to the center of the trap or be squished.

It should:

- Successfully allow the bug container at the bottom of the trap to be emptied out AND the container should hold at least 1000 SLFs.
 - The bugs need to be emptied out from the trap for it to have a long lifespan and to capture more bugs than its volume would otherwise allow.
 - If the container can't hold a significant amount of SLFs, it will need to be continuously emptied (unwanted labor burden)
- Allow more than 50% of the light from a given light source to pass through the clear top (measured using a smartphone light intensity meter)
 - Light needs to penetrate the trap in order to attract the bugs
- Successfully transport at least 6 SLF at one time from the top of the shaft to the bug-catching area (6 SLFs can fit between each section of ridges, so this is the maximum the tube will need to handle at once)
 - If multiple bugs land on the trap at once, all of the bugs should be able to be trapped
- Successfully squish a bug-shaped object that is on the plate while it rotates (result is binary: either it works (1) or it doesn't (0))
 - The bugs need to be either killed or forced to the center of the trap to work, so the ridges must establish the threat of death (to force the bugs into the tube or kill them as a backup)

Demo: We will likely demonstrate that it will squish a bug-shaped object that is on the plate while it rotates. We will place a small, approximately lanternfly-shaped object with flexible properties on the top table of the trap, then rotate it to show how the fly will be squished if it doesn't move to the center of the trap.